CSBB100

Project Report

on

VIDHYUT

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Table of Contents:

PROJECT APPROVAL....4
Declaration....5
ACKNOWLEDGEMENT....6
Chapter 1: Introduction....7
1.1 Problem description....7
1.2 Background....7
1.3 Organizing the Device....7
Chapter 2: Motivation....8
2.1 Industry Context and Trends....8
2.2 Market Demands....8
Chapter 3: Objectives....9
Chapter 4: Technology Used....10
4.1 OneLap GPS Module....10
4.2 GPS Integration....10
4.3 Backend and Data Management....10
Chapter 5: Methodology....11
5.1 Software Development....11
5.2 Hardware Integration....13
Chapter 6: Results....14
6.1 Future Scope....14
Chapter 7: Conclusion....15
Chapter 8: References....16

List of Figures

Figure	**Title**	Page No.
5.1.1	Environment setup for Flutter Framework	12
5.1.2	Environment setup on Visual Studio Code	12
5.1.3	Github Repo	12
5.2.1	Onelap GPS Module	13
5.2.2	Testing of Module in real-time	13

PROJECT APPROVAL

This Project work entitled " Vidhyut " by Harshit Jhamb , Gufran Alam , Kavitharkika Abhiram and Krish Karandikar approved for the degree of
Bachelor of Technology in Computer Science and Engineering

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Supervisor

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(Assistant Professor)

Date: 05th May 2025

Time: 11:00 am onwards

Place: National Institute of Technology, Delhi

Declaration

We declare that this written submission represents our ideas in our own words.
Where others' ideas or phrases have been included, we have adequately cited and referenced the sources. We also declare that we have adhered to all academic honesty and integrity principles and have not misrepresented, fabricated, or falsified any idea/data/fact/source in our submission. We understand that any above violation will cause disciplinary action by the Institute and can also evoke penal action from the authorities, which have thus not been properly cited, or from whom proper permission has not been taken when needed.

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Date: 05-05-2025

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First and foremost, we extend our deepest appreciation to our project guide, Dr Amandeep Kaur, for their constant support, expert guidance, and valuable suggestions at every stage of this project.

Sincere thanks also go to our fellow project teams. The healthy exchange of ideas, constructive discussions, and peer support played a significant role in refining our approach and enhancing the overall quality of our work.

This project has been a great learning experience, and we're grateful to everyone who played a part in making it a success.

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Chapter 1: Introduction

In today's rapidly evolving transportation landscape, the shift toward sustainable mobility has accelerated the adoption of electric vehicles (EVs) across the globe. With increasing numbers of EVs on the road, efficient fleet management and real-time tracking have emerged as critical needs to ensure both operational efficiency and safety. This project is driven by the imperative to modernize EV tracking through a system that combines advanced GPS technology, robust API communication, and an intuitive, cross-platform mobile application developed using Flutter.

1.1 Problem description

We are creating this project "Vidhyut" to track the real monitoring of the Electric Vehicle that saves the time of an individual and also works for the smooth regulation of the workflow. Furthermore, the presence of a cross-platform solution makes it accessible for diverse users.
This project addresses many issues by integrating advanced OneLap GPS, streamlined API communication, a Flutter-based mobile application and a Database for security purposes.

1.2 Background

The increasing global commitment to sustainable energy and reduced carbon emissions has accelerated the adoption of electric vehicles (EVs). As EVs become more common on our roads, managing these vehicles—whether in public transportation, ride-sharing fleets, or private ownership -- necessitates robust tracking and fleet management solutions to ensure security as well as efficiency of use.

1.3 Organizing the Device

Recent advancements in sensor and communication technologies have paved the way for more precise real-time tracking. The integration of state-of-the-art modules like OneLap GPS provides accurate geolocation even under challenging conditions, thereby addressing the shortcomings of earlier systems.

Chapter 2: Motivation

The primary objective behind this project is to bridge the gap between vehicle presence and user awareness by integrating GPS-based tracking with a user-friendly mobile application.This project not only enhances personal convenience but also addresses broader issues such as campus EV management, misuse prevention, and better resource allocation. The idea is to create a scalable, secure, and smart system that can eventually be adapted for larger fleets or public EV infrastructure.

2.1 Industry Context and Trends

Companies are investing in robust telematics solutions to offer better visibility and control over vehicle movement. For private EV users and institutions like colleges, however, affordable and lightweight tracking systems are still lacking.

This trend presents a unique opportunity for innovation in low-cost, scalable, and secure EV tracking solutions. The **Vidhyut project** aligns with these industry developments by proposing a practical system tailored to campus environments.

2.2 Market Demands

Despite the growth, there remains a gap in the availability of cost-effective, easy-to-use, and secure EV tracking solutions, especially for smaller markets like educational institutions, residential societies, and private users. Most commercial tracking systems are built for large-scale fleets and involve high recurring costs or complex installations that aren't suitable for everyday users. **Vidhyut** positions itself to address this market need by delivering a scalable solution that can evolve in step with the growing popularity and usage of electric vehicles.

Chapter 3: Objectives

The primary objective of the *Vidyut* project is to design and develop a smart, real-time electric vehicle (EV) tracking system that enhances the overall efficiency of fleet management. By integrating advanced GPS technology, real-time data communication, and a user-friendly mobile application, the system aims to address common challenges in EV fleet operations.

The key aims of the project include:

1. Time Optimization: Enable efficient route planning and reduce unnecessary travel time through accurate real-time tracking and updates.
2. Cost Reduction: Minimize operational costs related to fuel (in hybrid models), maintenance, and manual monitoring by automating tracking and reporting processes
3. Scalability and Flexibility: Build a system architecture that can easily scale to accommodate larger fleets and integrate with emerging technologies.
4. Secure Data Handling: Use Firebase as a secure backend to ensure the privacy and integrity of sensitive tracking data.
5. Cross-Platform Accessibility: Provide a seamless experience for users on both Android and iOS platforms via a single mobile application.

Chapter 4: Technology Used

This project integrates multiple modern technologies to provide a real-time, secure, and user-friendly electric vehicle tracking solution. Below are the key technologies used in the development of the system:

4.1 OneLap GPS Module

At the core of the system is the OneLap GPS module, a high-precision GPS solution designed for real-time tracking. This module provides accurate geolocation data, enabling continuous monitoring of vehicle positions under various conditions. The OneLap GPS module is integrated seamlessly into the system, ensuring reliable data acquisition that is critical for effective navigation. Its great performance enhances the overall efficiency of the EV tracking system.

4.2 GPS Integration

Accurate geolocation is critical for an effective EV tracking system. The project leverages GPS integration to capture precise, real-time location data of electric vehicles. This technology allows continuous monitoring of vehicle movements, ensuring that users receive timely and dependable updates essential for navigation, route planning, and fleet management.

4.3 Backend and Data Management

To secure and manage the sensitive location data generated by the GPS system, Firebase Database is employed as the backend solution. Firebase not only ensures real-time data synchronization but also provides robust security features such as user authentication and data encryption. This combination guarantees that all records are stored securely while remaining easily accessible to authorized users, maintaining both performance and privacy.

Chapter 5: Methodology

5.1 Software Development

The software component of the *Vidhyut* project focuses on creating a reliable and user-friendly mobile application for tracking electric vehicles in real time.

Development Tools and Environment:

1. Flutter Framework: Used for developing a cross-platform mobile application using a single codebase for both Android and iOS.

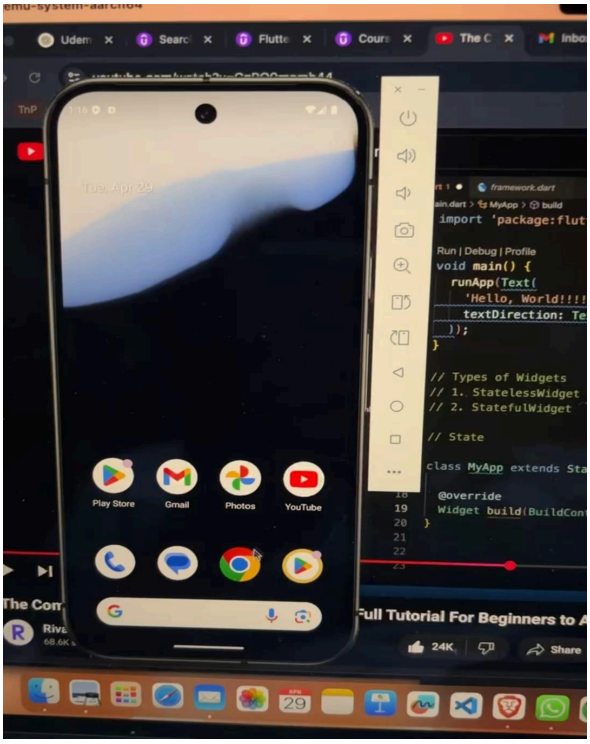


Fig 5.1.1

2. Dart Programming Language: The primary language used with Flutter to build the application logic and UI.

3. Visual Studio Code (VS Code): Chosen as the primary IDE due to its extensive Flutter/Dart support, ease of use, and integration with version control tools.

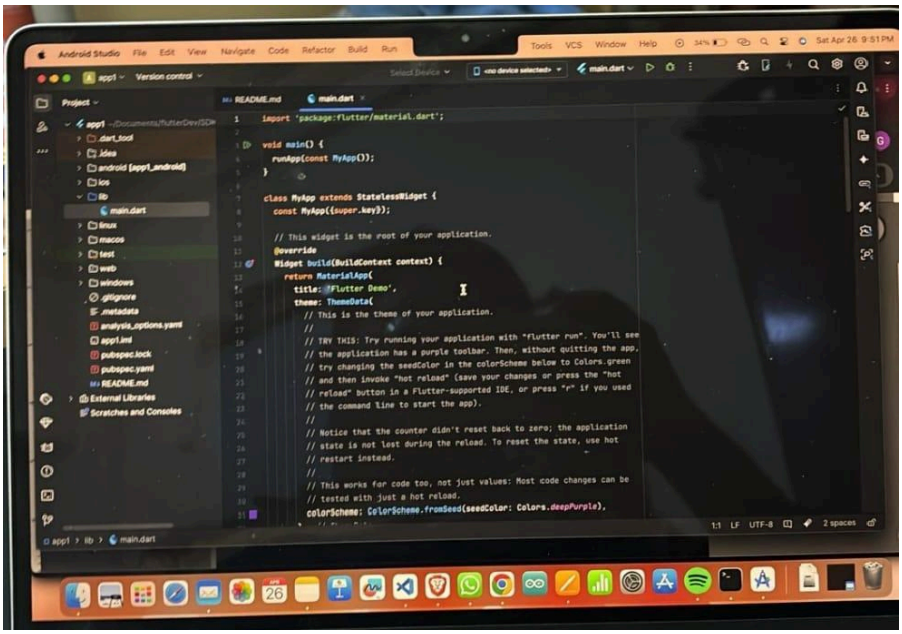


Fig 5.1.2

4. GitHub:
   Visit the below site for more information regarding EV_tracker_project_Vidhyut
   https://github.com/Abhiram1103/EV_Tracker_Vidyut.git

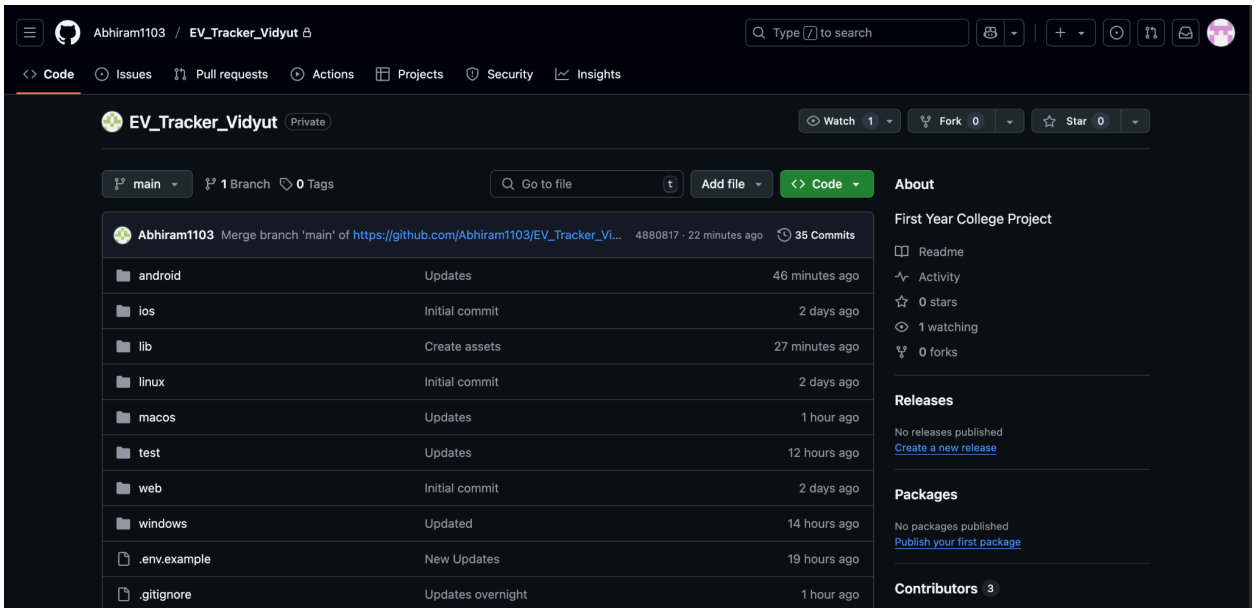


Fig 5.1.3

5.2 Hardware Integration

Onelap GPS Tracking Device:

A compact and commercial-grade GPS tracker used to retrieve real-time geographic coordinates Installed inside the EV and powered by the vehicle's battery or internal power supply.



Fig 5.2.1

The GPS device was securely mounted inside the EV and tested across multiple locations within the campus to ensure signal strength and accuracy.

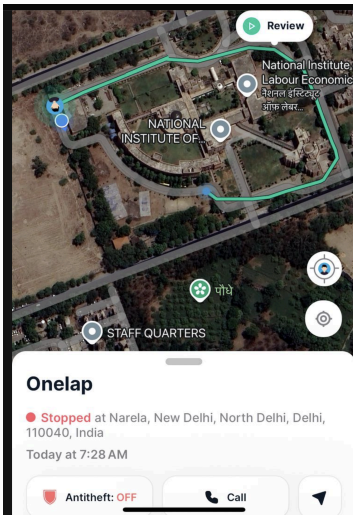


Fig 5.2.2

Chapter 6: Results

The *Vidhyut – EV Tracer* project successfully demonstrated the potential of building a smart, secure, and scalable EV tracking system using affordable hardware and modern app development technologies. Beyond basic tracking, the system showed promise for real-world usability and future smart campus integrations.

1. Real-Time Tracking with High Accuracy:
Achieved consistent location accuracy within ±5 meters, based on GPS signal and environment. The system was able to track EV movement across the entire campus without disconnection.
2. Performance on Different Devices
The app was tested on multiple Android phones and one iPhone.It maintained consistent UI performance and responsiveness across devices with different hardware capabilities.
3. Security and Access Control
Firebase ensured secure data access and updates. A custom rule was implemented to allow only authenticated users to read/write location data, preventing unauthorized tracking.
4. Time Management
This system enables smart time management for users by allowing them to locate electric vehicles (EVs) in real time. In campus environments, where several EVs are used, users often waste time searching for available or nearby vehicles. The Vidhyut system solves this challenge

6.1 Future Scope

1. Multi-EV Tracking System
   In future iterations, the system can be extended to track multiple EVs simultaneously. Each EV will have a unique ID in the Firebase database, allowing the app to switch views or show all available EVs on a single map interface.
2. Headcount Detection in EVs
   - Using sensors (e.g., weight sensors, infrared counters, or basic camera-based analytics), the system can estimate or count the number of people currently in the EV.
   - This data can be used to:
   - Prevent overcrowding.
   - Suggest alternative EVs if one is full.
   - Collect usage statistics for optimizing future deployments.
3. Availability-Based Notifications
   The module will send the notification whether the EV is in active state or in charging state.
4. Reservation and Booking System
   Users could reserve EVs in advance through the app.

Chapter 7: Conclusion

The EV Tracker System project has proven to be a practical solution for tackling common issues in fleet management and vehicle tracking. By integrating advanced tools like the OneLap GPS module, the system delivers precise and consistent location data—essential for planning routes efficiently and ensuring driver and vehicle safety. Additionally, the project features smooth communication between hardware components, backend services, and the user interface, thanks to well-structured API connections that keep real-time data flowing without interruption.

Building the mobile app using Flutter allowed us to deliver a smooth, cross-platform experience that works seamlessly on both Android and iOS devices. This not only improved usability for end users but also made ongoing updates and maintenance more efficient for developers. On the backend, we chose Firebase Database to securely manage tracking data, ensuring strong data protection and helping build user trust in the system.

From the start, we focused on identifying and addressing potential performance bottlenecks, with scalability in mind. The system architecture was built to accommodate future features, whether it's adding predictive analytics, more sophisticated geofencing, or integration with smart city infrastructure. This future-ready design reflects our goal of supporting both current operational needs and long-term progress in sustainable transport solution

In summary, this project takes a well-rounded approach by combining accurate GPS tracking, secure real-time data handling, and a user-friendly mobile app. Together, these elements create a solid foundation for future advancements in EV fleet management. The system effectively tackles current pain points while paving the way for innovations that could significantly transform how electric vehicle fleets are tracked, maintained, and optimized.
In this way, we have successfully created our project "Vidhyut".

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    1. Sudhakar, A. & Kumar, B.. (2024). EV Fleet Energy Management Strategy For Smart Microgrids Considering Multiple Objectives: Techno-Economic Perspective. Arabian Journal for Science and Engineering. 49. 10.1007/s13369-024-09209-w. This study presents a comprehensive energy management strategy for EV fleets within smart microgrids, addressing multiple objectives from a techno-economic perspective. It explores the integration of fuzzy controllers to optimize battery degradation and energy efficiency in EV fleets.
    2. Mariprasath, T. & Venkatanaraya, M.. (2024). Design and development of new charge controller for GPS tracker for electric vehicles. Discover Electronics. 1. 10.1007/s44291-024-00011-4. This research introduces a novel charge controller designed for GPS trackers in EVs, emphasizing sustainable power sources like solar PV systems to ensure continuous and efficient tracking. The study validates the system's capacity to deliver dependable power, enhancing remote monitoring and fleet management.